

Joint inversion of TEM and distorted MT soundings: Some effective practical considerations

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ABSTRACT

Accurate layered-earth interpretation of magnetotelluric (MT) apparent resistivity sounding curves distorted by the presence of near-surface, 3-D bodies in an otherwise horizontally stratified ground requires that the curves be shifted to the level they would have assumed in the absence of these anomalous small-size bodies. The most popular preinterpretation processing involves shifting the biased MT apparent resistivity curves to the levels determined by some additional data. This paper examines the nature of TEM field data, their relationship with MT data, and their usefulness in constraining the interpretation of MT data at locations affected by near-surface 3-D effects. The MT apparent resistivity curves are neglected in a joint inversion scheme involving the (undistorted equivalent) phase information and TEM data, thus eliminating the need for preinterpretation data processing or correction of statically shifted curves. This approach is efficient and effective, and since an overlapping TEM observational bandwidth is not always required for a successful joint interpretation, simple inexpensive TEM equipment can be deployed as a logical complement to MT in deep basin exploration campaigns.

INTRODUCTION

Field and theoretical studies have shown that in the presence of small-scale surface or near-surface inhomogeneities, magnetotelluric (MT) apparent resistivity sounding curves are typically shifted by a multiplicative factor which, for a given measurement direction, is constant at all frequencies with the impedance phase data being unaffected (see e.g., Jones, 1988 and references therein). MT static shift, as it is termed, is essentially a frequency-independent galvanic

response associated with 3-D bodies (e.g., Berdichevsky and Dmitriev, 1976), and the direction and possibly the amount of shift depends on the resistivity characteristics of the 3-D bodies. One of the problems faced in conventional MT data analysis is the 2-D nature of the typical survey plan, which means the exact forms and locations of the near-surface 3-D bodies are unknown. Consequently, such heterogeneities cannot be realistically included in a 3-D interpretation scheme (e.g., Wannamaker et al., 1984a, 1984b), and conventional wisdom is to seek some efficient corrective measures for an otherwise ill-defined problem.

For conventional linear surveys, an approach to solving the static shift problem in the absence of any complimentary information is to shift the suspected apparent resistivity sounding curve to some reference level determined with the aid of some statistical identification or characterization criteria (e.g., Jones, 1988; Beamish and Travassos, 1992). A more unique extension of this practice incorporates a priori information from complimentary surface measurements (e.g., Andrieux and Wightman, 1984; Sternberg et al., 1988; Pellerin and Hohmann, 1990; Meju and Fontes, 1993). Note that borehole induction logs could, in principle, furnish control data (formation thicknesses and resistivities) for static shift assessment, but as noted by Beamish and Travassos (1992), some discrepancies may be found in the resistivity values determined from well logging (which measure the properties only in the immediate vicinity of the well) and those determined from MT soundings (which average over large volumes of the subsurface) (see also Orange, 1989). In any case, even if there is a perfect correlation between borehole and MT data, drawbacks in the use of borehole resistivity-depth information in static shift correction are that access to borehole logs is limited in many routine surveys and that geological formations typically show some form of lateral variations in thickness, degree of consolidation, and fluid content such that interpretations based on the data furnished by a few boreholes may not always be accurate.

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The data correction and interpretation approach that uses independent surface electromagnetic data derives essentially from the benchmark work of Larsen (1977) which suggested that galvanic distortions (i.e., those affecting only electric fields) can be corrected by using information provided by a complimentary induction method that uses only magnetic fields for subsurface characterization. The transient electromagnetic (TEM) method is becoming increasingly popular as a complimentary tool to MT in deep subsurface resistivity investigations, and remarkable work on the correction of statically shifted MT apparent resistivity curves has been done by Sternberg et al. (1988) and Pellerin and Hohmann (1990) using central loop TEM apparent resistivity data. It may be noted, however, that these TEM workers focused on the correction and interpretation of the MT apparent resistivity data (a two-stage data analysis process, *sensu stricto*) even though the impedance phase, an equivalent data set, is not affected by static shift. In any case, Sternberg et al. (1988) demonstrated the interpretational accuracy of the corrected MT curves by comparison with well log information, and Pellerin and Hohmann (1990) highlighted the importance of TEM-based corrections in multidimensional environments.

One limitation of the TEM-based procedures suggested in these landmark papers is that an overlap between (preferably flat segments of) the TEM and MT sounding curves is necessary for the techniques to be effective, which requires TEM soundings over a considerable bandwidth. Deep probing wideband TEM instrumentation is expensive and often logistically cumbersome (in comparison with some of the common natural source MT field systems). However, relatively inexpensive field portable TEM equipment is now available for shallow-depth probing (see e.g., Meju, 1994a) but may not furnish overlapping data with the typical deep-probing MT systems used in basin exploration. Also, in some TEM poor-record areas (such as highly resistive environments) only a portion of the recorded time windows may be useable and may not overlap with the MT data window.

This paper advocates the joint inversion of MT-phase and TEM apparent resistivity data as a more versatile extension of the previous work on TEM-based analysis of distorted MT sounding data. Since no corrections to the distorted MT apparent resistivity curves are necessary in this approach, the scheme could also be regarded as a cost-effective one-stage alternative to the dual-stage interpretational process of apparent resistivity shifting and inversion. It will be shown that for typical exploration problems, the coincident loop TEM method can be reliably employed for static shift remediation and that an overlap between the TEM and MT data sets is not always essential for retrieving a reliable subsurface resistivity structure. Consequently, a simple inexpensive TEM system may be deployed as a logical complement to MT in routine surveys. I will first assess the reliability of the TEM field data to be used for constraining the MT interpretations.

NATURE OF TEM FIELD DATA

It is instructive to gauge the potency or reliability of TEM soundings in different geotectonic environments before prescribing them as an elixir for the MT static shift problem.

Joint data sets from several (over 140) coincidentally located MT and TEM surveys for hydrocarbons, groundwater, and mineral deposits in the Irish Caledonides, the Trans-Iberian Pyrite Belt, the Troodos Ophiolite Complex of Cyprus, and the Parnaíba Basin in northern Brazil (e.g. Meju, 1994b; Meju and Fontes, 1993; Meju et al., 1993) have been analyzed by this author for consistency or behavioral patterns. In these surveys, TEM soundings were made using rectangular transmitter (Tx) loops of the same side length as (and in line with) the MT electric dipoles (to ensure that both methods are sampling essentially the same geology), and if permitted by the available field system, both the central loop and coincident loop TEM data were recorded at the same station (to ascertain possible structural differences in the vicinity of the Tx loop). Also, some of these surveys involved grid-type measurements with station spacings of 100 m or less for good resolution of possible short-wavelength features where the targets were 3-D massive sulphide bodies concealed at depth (and static shift patterns are sought as possible target discrimination criteria). Some of the noted response characteristics are briefly discussed next.

Analysis of over 140 coincident TEM and MT field soundings and numerical modeling studies suggest that the MT wave frequencies f are related to transient times t in the form (cf. Sternberg et al., 1988)

$$t = \frac{1}{3.9f}, \quad (1)$$

where t is in seconds and f is in hertz and provides a basis for the comparative analysis of TEM and MT soundings in this paper. It was also observed that for stratified geological environments, the recorded TEM apparent resistivity ρ_a is related to the effective subsurface resistivity in the form

$$\rho_{\text{eff}} = k\rho_a e^{-(1-\alpha)} \quad (2)$$

and distributed at a depth approximated by

$$\delta_{\text{eff}} = [(2t\rho_a/\mu_0)^{1/2}]/k, \quad (3)$$

where the parameter α in the exponent is assigned a value of 0.15, the constant k has a value of 2.3, t is the sampling time in seconds, and μ_0 is the magnetic permeability of free-space [see Meju (1995) for a generalized transformation scheme]. These expressions fit field data from horizontally stratified environments best (see Meju, 1994a), and hence the approximate resistivity-depth characteristics of a given TEM data set may serve as a crude dimensionality indicator (e.g. Meju, 1994a). Note that all the respective TEM data are routinely adjusted for system turn-off effects (e.g. Raiche, 1984) in an iterative apparent resistivity determination process.

In a dominantly horizontally stratified (i.e., 1-D) geological environment, the TEM and MT methods would be expected to furnish equivalent apparent resistivity curves. A comparison of the TEM data and two orthogonal MT data from the same location in the Parnaíba Basin (Meju et al., 1993) is given in Figure 1. The MT data show 1-D features at frequencies above 1 Hz and will serve to demonstrate the similarity of TEM soundings. Notice the good match between the overlapping steep segments of the three apparent resistivity curves; one of the MT polarizations has been slightly affected by static shift. The popular Bostick (1977)

transformations of the MT data are also compared with the TEM resistivity-depth transformation based on equations (1) and (2) to demonstrate the agreement between the respective interpretive models in 1-D environments.

To assess the reliability and the response characteristics of TEM in a simple 2-D geological setting, overlapping broadband data sets from a hydrocarbon exploration campaign in Northern Ireland have been chosen, and sample joint data are compared in Figure 2. The TEM data were acquired using the deep-probing Geonics EM42 system with a Tx loop of 300 m side length, while the MT data were recorded using the S.P.A.M. Mk2 system developed by Edinburgh University. The dominant geological strike at this location is northeast-southwest (about N45°E) and is related to the Caledonian orogeny of about 500 million years ago. Although the MT impedance rotation angles may have a 90° ambiguity, the calculated azimuths for the major impedance were practically invariant with the observational frequencies and have values of about N48°E (magnetic), which are consistent with the known geological strike. The good match between the broadband TEM and MT minor apparent resistivity curves can be seen in the left-hand panels in Figure 2. Note that the major curve, the unrotated curves (shown in the right-hand panels in Figure 2), and the rotationally invariant data (not shown) do not provide a close match at the TEM observational bandwidth at this station. It is instructive to evaluate the utility of 1-D inversion [based on equations (1) and (2)] in such a 2-D environment. Notice in

Figure 3 that the 1-D interpretation of the TEM data yields a model whose response satisfactorily fits the apparent resistivity data. However, the computed MT phase response for this model does not replicate the observed phase data for the matching MT polarization. It is also pertinent to mention here that the broadband TEM data matched the major resistivity best at other sites along the Irish transect, thus a careful selection of the interpretive parameters is imperative in such environments.

In an effort to evaluate the field responses in a true 3-D setting, joint soundings were made over known massive sulphide bodies. Note that small TEM loops and MT electric dipole lengths (25 m–50 m) were necessary in these surveys. A sample data set from a survey over the Klirou deposit in Cyprus is shown in Figure 4. The sulphide target lies at a depth of about 80 m to 110 m within geochemically altered pillow lavas and is areally restricted (G. Maliotis, private comm., 1991). The MT electric dipoles were located within the projected surface position of the ore body. Notice that the TEM apparent resistivity data lacked diagnostic signatures that may serve to delineate the target (see also Cooper and Swift, 1994). It would appear that the prevalent geochemical alterations (silicification) in this area have rendered the massive conductor indistinguishable from the host pillow lavas. In any case, the MT data over this deep body appear to have been affected by undesirable near-surface distortions. The computed MT azimuths show two trends that agree with the strikes of the faults associated with the

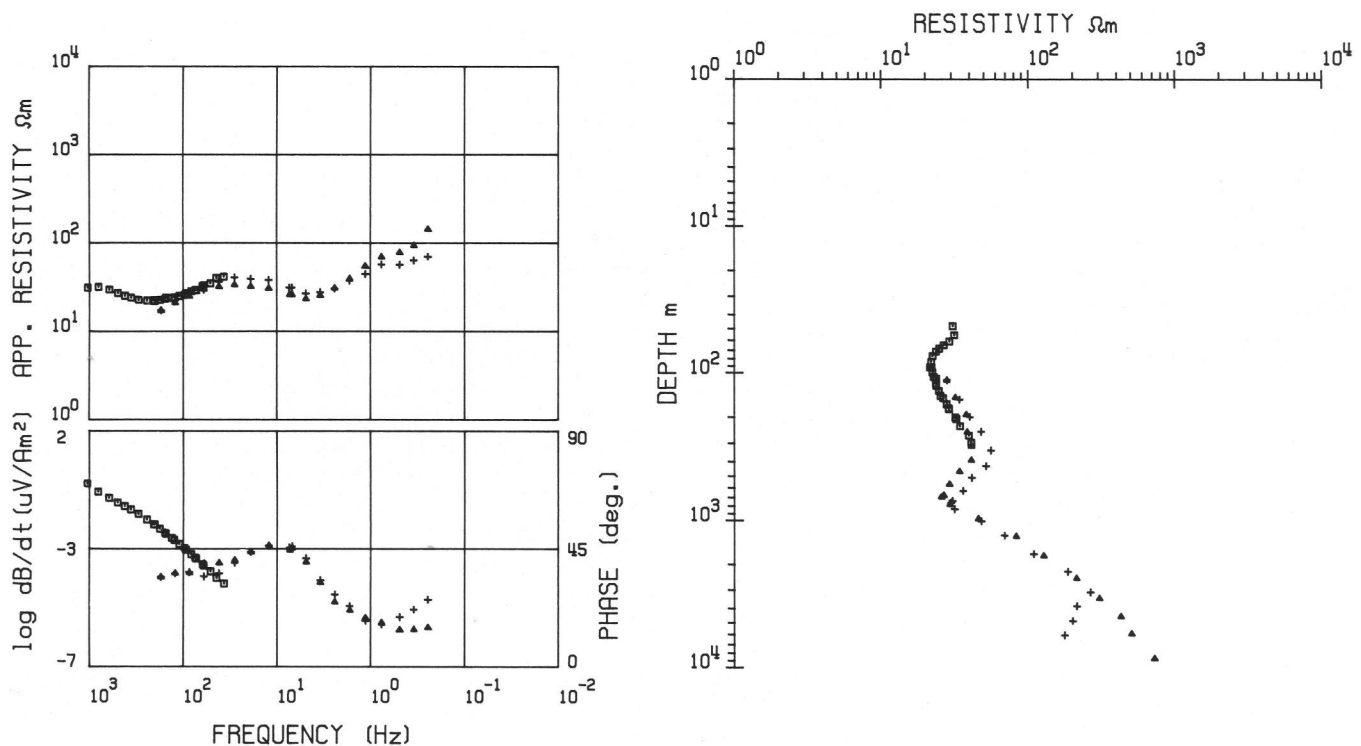


FIG. 1. A comparative analysis of TEM and dual polarization MT data from a station with 1-D shallow-depth conditions. On the left-hand panels are shown the apparent resistivity data (top plot) and the MT phase and TEM voltage decay curves (bottom panel). The TEM data are shown by squares, and the MT xy and yx polarizations are represented by the triangles and crosses, respectively. The Bostick (1977) transformations of the MT data are shown together with the similarly transformed TEM data on the right-hand plot.

deposit, but it could be seen that the TEM apparent resistivity curve is somewhat parallel to one of the MT polarization curves at the overlap region. However, the other MT polarization data may pose a difficult problem, especially as the position of the sulphide mass cannot be accurately determined using only the TEM data. This is a common response pattern in these surveys.

The TEM field examples shown in Figures 1 to 4 are central loop data. The central loop technique is commonly chosen for MT static shift corrections since the strength of the induced signal is highest at the center of the Tx loop. However, experience has shown that for such tasks, and depending on the sounding bandwidth, the logistically simpler coincident loop technique (the same wire loop serves as the transmitter and receiver) generally yields equivalent information, especially in areas of simple geology. In Figure 5 both the central and coincident loop TEM apparent resistivity data and two orthogonal MT polarization data are shown for sample locations at the southeastern margin of Parnaíba Basin. Notice the agreement between the central and coincident loop data in these examples. The Sirotem Mk2SE TEM equipment (powered by a 24V battery source) and the EMI MT system were used in these soundings. The gross structural framework in the region is 2-D (there is a general dip away from the basin margin but the geological stratification is interrupted by occasional basaltic dyke and sill injections). It is obvious that either of the central or coincident loop data sets would suffice for static shift remediation operations at these stations. Let us now attempt to

reconcile the above field observations with a feasible inversion scheme.

REGULARIZED INVERSION METHOD

The inverse problem is to retrieve a representative model of the subsurface from a collection of complimentary data from two or more sources at specified surface observational stations. Joint inversion of geophysical data is not new (see e.g. Vozoff and Jupp, 1975; Gol'stman, 1976), but there may be significant differences in the derivative regularization measures employed in each scheme. The geoelectromagnetic induction problem is nonlinear and of the form

$$\mathbf{d} = f(\mathbf{m}) + \mathbf{e}, \quad (4)$$

where the field observations are described by the set of numerical values contained in the vector $\mathbf{d}$, $f(\mathbf{m})$ is the theoretically predicted observable response to inductive energization of our hypothetical earth-model represented by the set of parameters $\mathbf{m}$ (here, the resistivities and thicknesses of the various layers or blocks in our earth-model), and $\mathbf{e}$ is the vector of additive noise in our observations.

For the case involving a single method, we can linearize the problem around an initial model $\mathbf{m}^0$ using Taylor's theorem in the usual manner. In the regularized inversion procedure adopted here, the objective is to minimize the function

$$L = (\mathbf{d} - f(\mathbf{m}))^T (\mathbf{d} - f(\mathbf{m})) + \beta (\mathbf{m}^T \mathbf{D}^T \mathbf{D} \mathbf{m}), \quad (5)$$

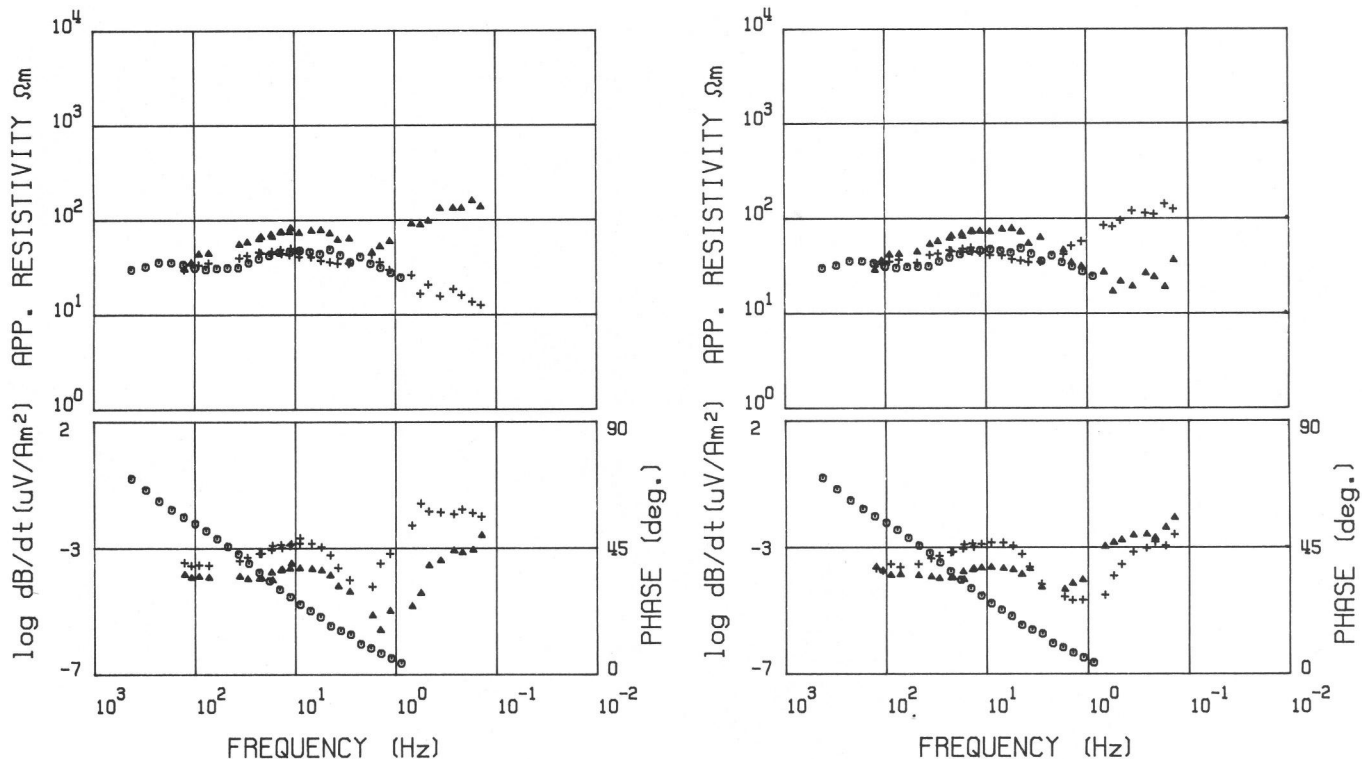


FIG. 2. A comparison of TEM and MT data sets from a 2-D environment. The rotated (major and minor) and unrotated (xy and yx) MT data are shown together with the TEM data (open circles) on the left-hand and right-hand panels, respectively. The major and xy data are represented by triangular symbols while the minor and yx data are shown as crosses.

or the linearized equivalent

$$L = (\mathbf{y} - \mathbf{A}\mathbf{x})^T(\mathbf{y} - \mathbf{A}\mathbf{x}) + \beta(\mathbf{m}^0 + \mathbf{x})^T \mathbf{D}^T \mathbf{D}(\mathbf{m}^0 + \mathbf{x}), \quad (6)$$

where $\mathbf{y} = \mathbf{d} - f(\mathbf{m}^0)$ is the discrepancy vector, $\mathbf{x} = (\mathbf{m} - \mathbf{m}^0)$ is the vector containing the sought corrections for $\mathbf{m}^0$, $\mathbf{A} = \partial f(\mathbf{m}^0)/\partial \mathbf{m}^0$ is the matrix of partial derivatives, $\mathbf{D}$ may be the identity matrix or a first difference operator, β is an undetermined multiplier, and the superscript T denotes the transposition.

Minimization is effected as usual by setting to zero the derivatives of L with respect to $\mathbf{x}$ and solving for $\mathbf{x}$. The constrained least-squares solution is given by (e.g., Meju, 1994c, 1994d)

$$\mathbf{x} = [\mathbf{A}^T \mathbf{A} + \beta \mathbf{H}]^{-1} \{\mathbf{A}^T \mathbf{y} - \beta \mathbf{H} \mathbf{m}^0\}, \quad (7)$$

where the matrix $\mathbf{H} = \mathbf{D}^T \mathbf{D}$.

For the more relevant joint inversion problem, the above formula expands to

$$\mathbf{x} = [\mathbf{A}_*^T \mathbf{A}_* + \beta \mathbf{H}]^{-1} \{\mathbf{A}_*^T \mathbf{y}_* - \beta \mathbf{H} \mathbf{m}^0\}, \quad (8)$$

where

$$\mathbf{y}_* = \begin{bmatrix} y_t \\ y_\phi \end{bmatrix} = \begin{bmatrix} d_t - f_t(\mathbf{m}^0) \\ d_\phi - f_\phi(\mathbf{m}^0) \end{bmatrix} \quad (\text{dimension: } [nt + n\phi] \times 1)$$

and

$$\mathbf{A}_* = \begin{bmatrix} \mathbf{A}_t \\ \mathbf{A}_\phi \end{bmatrix} = \begin{bmatrix} \partial f_t(\mathbf{m}^0)/\partial \mathbf{m}^0 \\ \partial f_\phi(\mathbf{m}^0)/\partial \mathbf{m}^0 \end{bmatrix}$$

(dimension: $[nt + n\phi] \times np$)

for the nt (≥ 1) TEM apparent resistivity data, the $n\phi$ MT phase measurements, and the np model parameters. The subscripts t and ϕ denote the contributions to the system of equations from TEM and MT phase data, respectively.

Some effective practical considerations

It is good practice in inversion to scale each observation by its associated uncertainty so that poorly estimated data have relatively less influence on the solution. In this situation, the weighted solution is

$$\mathbf{x} = [(\mathbf{W}\mathbf{A})_*^T (\mathbf{W}\mathbf{A})_* + \beta \mathbf{H}]^{-1} \{(\mathbf{W}\mathbf{A})_*^T (\mathbf{W}\mathbf{y})_* - \beta \mathbf{H} \mathbf{m}^0\}, \quad (9)$$

where the diagonal weighting matrix $\mathbf{W}$ contains the reciprocals of the observational errors. The above parameter corrections $\mathbf{x}$ are applied to $\mathbf{m}^0$ in successive steps to recover the desired model parameters $\mathbf{m}$. It may be noted, however, that the two sets of data to be inverted have different magnitudes. It is therefore instructive to normalize them to some common-scale before they are assembled in $\mathbf{y}_*$ and $\mathbf{A}_*$. Now, as in conventional simultaneous inversion of MT apparent resistivity and phase data (e.g., Meju 1992), the

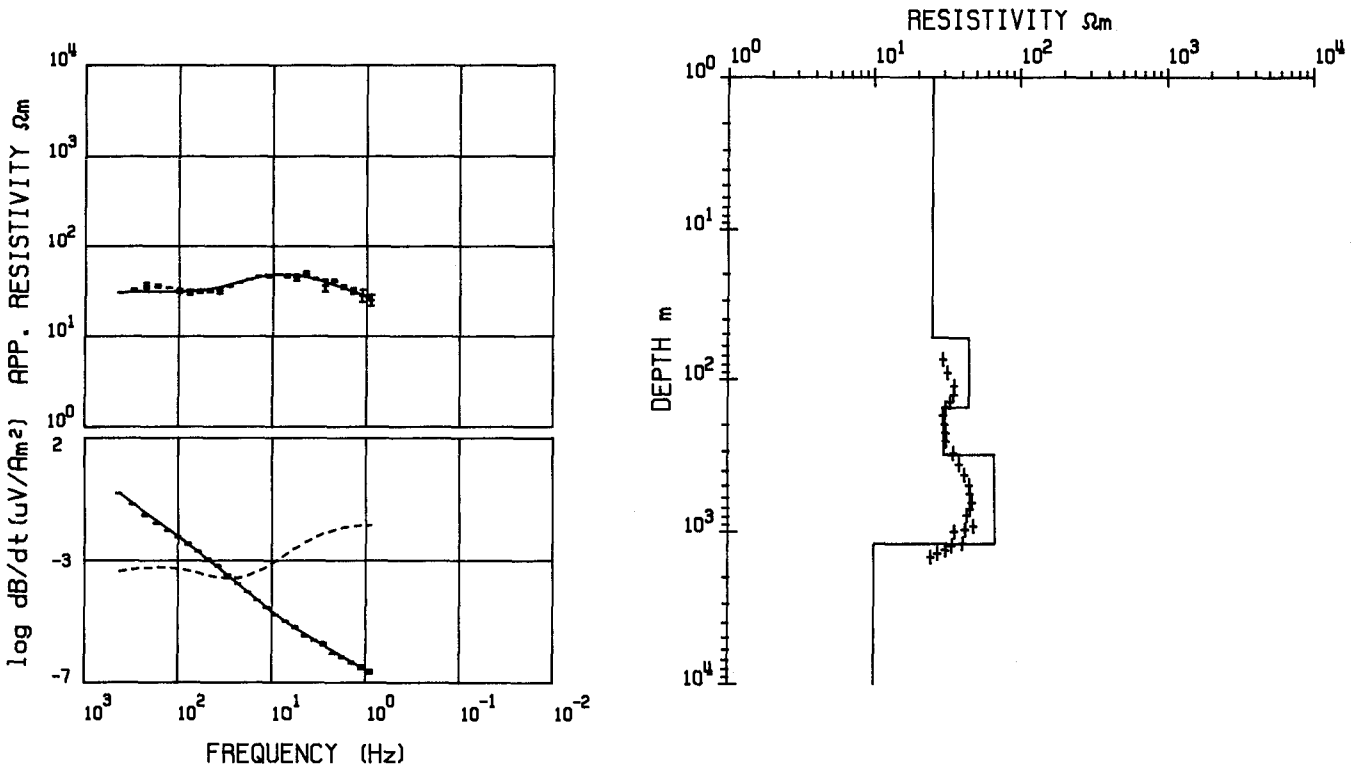


FIG. 3. Result of 1-D approximate inversion of TEM data over a 2-D structure. On the right-hand plot are shown the blocked 1-D structure corresponding to the simple data transform (shown as crosses). The computed TEM responses (solid curves) of this model are superimposed on the field data (shown with associated standard errors) on the left-hand panels. The computed MT phase data for this TEM model is also shown on the left-hand plot (dashed curve in bottom panel).

logarithms of the TEM data are considered in this scheme with the actual phase data, i.e.,

$$\mathbf{y}_t = \{\ln \mathbf{d}_t - \ln f_t(\mathbf{m}^0)\}$$

and

$$\mathbf{A}_t = \partial\{\ln f_t(\mathbf{m}^0)\}/\partial \mathbf{m}^0.$$

The components of $\mathbf{m}$ are taken to be the logarithms of the resistivities and thicknesses of the subsurface model sought. However, since the phase data could vary from 0 to 90° while the transformed apparent resistivities would typically assume a range of values that is less than one-tenth of that of the phase data, a useful practical strategy might be to normalize the contributions from the phase data to the system of equations by some constant factor (< 10). This latter strategy is not necessary for data acquired in areas characterized by strong horizontal geological stratification (i.e., 1-D environments).

APPLICATIONS TO FIELD DATA

1-D geological environments

If horizontal stratification is the predominant geological feature of an area, it could be expected that the MT phase data will be similar in any two orthogonal measurement directions such that a representative model for a given observational station may be obtained by inverting the combined phase data sets. (Note that the rotationally invariant MT phase may be selected if we are interested in

interpreting a single set of phase data.) Applications of this “combined phase” inversion strategy to a set of coincidentally located TEM and MT depth soundings that show 1-D characteristics (at frequencies greater than 1 Hz) are shown in Figure 6. The MT data sets have been affected by different amounts of static shift. The result of inverting the combined phases and the full set of overlapping TEM data is shown in Figure 6a while the inversion result incorporating only a portion of the TEM data (to simulate a band-limited TEM data set or nonoverlapping situation) is shown in Figure 6b for comparison. Notice the similarity between the two independently derived optimal joint models and the good fit of the model responses to the MT-phase and TEM data in both cases. The general parallelism of the computed MT apparent resistivity curves to the actual field data is also noteworthy; the conventional schemes would have had to shift the apparent resistivity field curves upwards by different amounts to retrieve these concordant optimal subsurface resistivity distributions. The geological stratification at this sounding location consists of a dipping cyclic sequence of marine shales and sandstones, and Meju et al. (1993) have already shown the agreement between an independently derived TEM-MT regional model and borehole and geological data.

Another useful feature of simultaneous MT-phase and TEM apparent resistivity data inversion is perhaps the fact that TEM data of limited bandwidth would suffice for recovering the true, or at least a reliable, subsurface resistivity structure from the MT phase data in areas of simple geology.

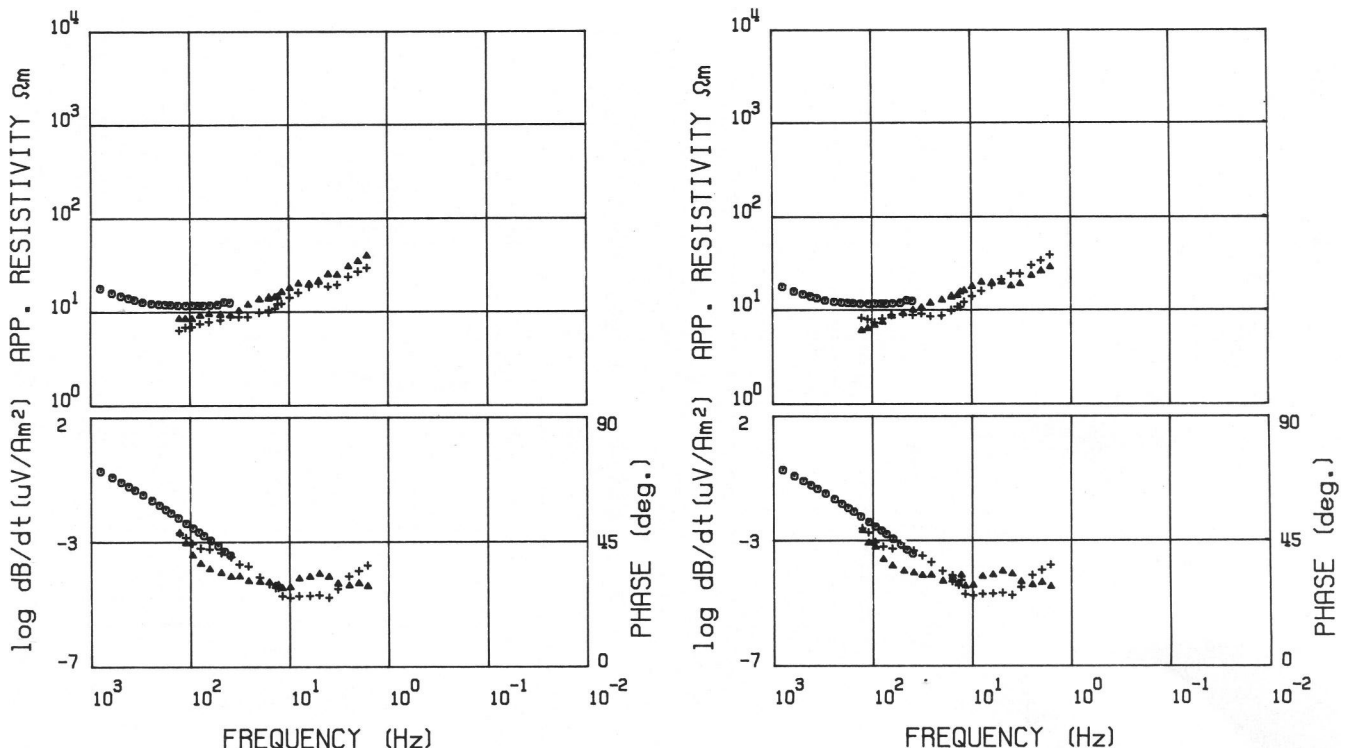


FIG. 4. TEM and MT data sets recorded over a concealed 3-D massive sulphide deposit. The data types and symbols are the same as for Figure 2.

In fact, a single (preferably overlapping) complimentary datum would suffice for some joint interpretation problems. This approach may therefore lead to considerable savings in measurement and computational times for judiciously chosen TEM time windows in joint basin exploration campaigns.

Multidimensional geological environments

In areas of complex geology, the suggested direct interpretation strategies would hold if the inversion scheme incorporates a forward routine appropriate to the subsurface structural dimensions. In other words, a 2-D MT simultaneous dual polarization phase inversion with TEM constraints would solve the problem of static distortions in an area where there is a dominant regional strike (i.e., a 2-D structure). In any case, if one is irrevocably committed to correcting the MT apparent resistivity data by shifting the curves to their correct levels as in conventional practice, then it may be pointed out that the appropriate sets of orthogonal polarization data may be uniquely corrected only

if 1-D conditions prevail above the anomalous structure and the TEM data are at a sufficiently high (equivalent) frequency to ensure true 1-D response. Preliminary modeling studies of field response patterns of the type shown in Figure 2 appear to suggest that, provided the appropriate pair of MT polarizations are selected for simultaneous inversion, contiguous or overlapping TEM apparent resistivity data for the matching MT polarization can furnish reliable constraints on the combined (i.e., dual polarization) phase data inversion process. It remains to assess the usefulness of the proposed scheme in a 3-D earth.

To a great extent the situation represented by the Klirou prospect in Cyprus evokes a picture of an approximately layered ophiolitic terrane with an embedded 3-D ore body. Although the TEM data cannot pinpoint the position of the ore body, they can distinguish between the barren upper pillow lava and the altered lower pillow lava hosting the sulphide mineralization (see also Cooper and Swift, 1994). It is expected that they will furnish reliable estimates of the average resistivities and thicknesses of the lavas for MT

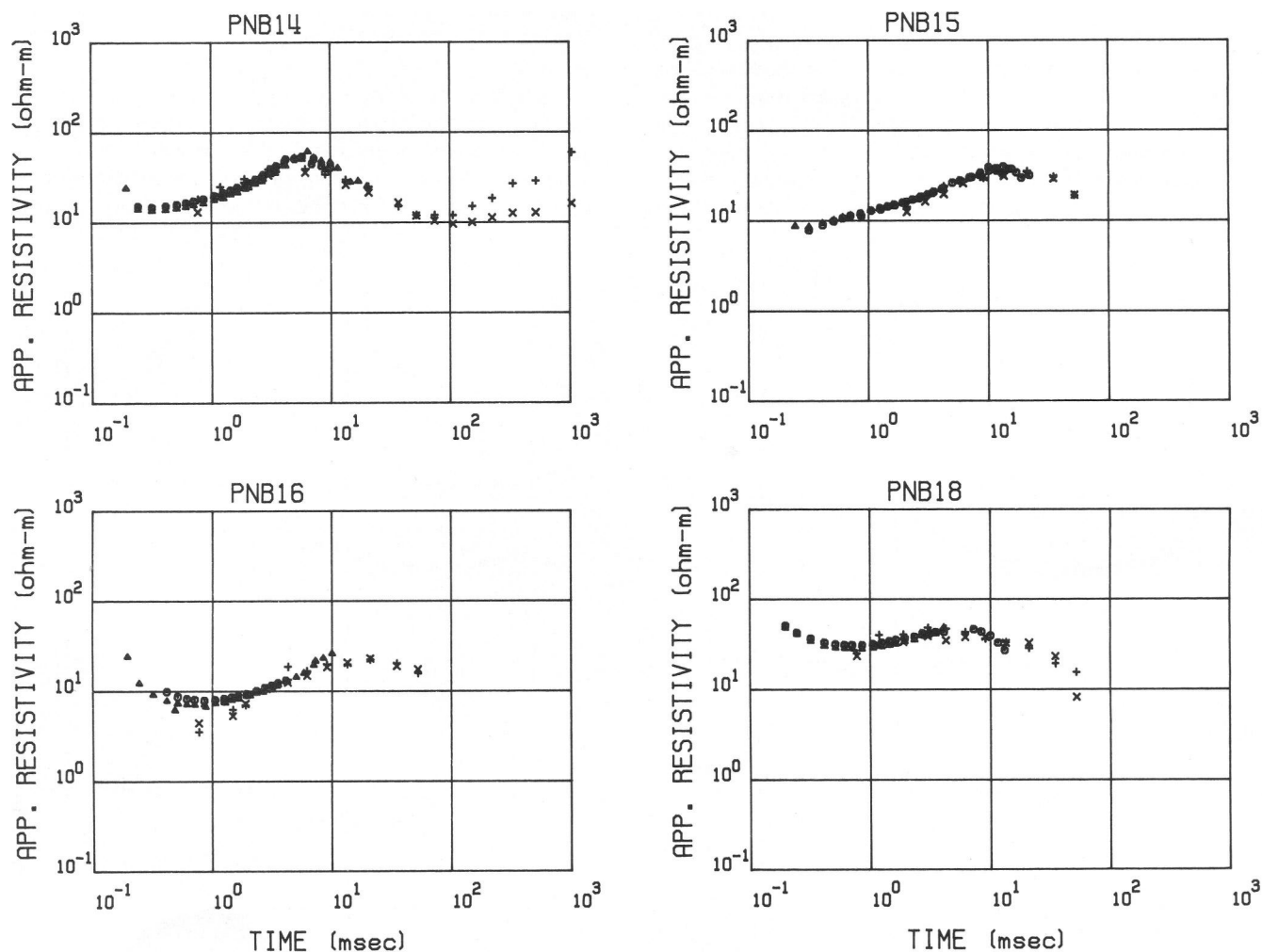


FIG. 5. Comparison of central loop and coincident loop TEM depth soundings in a sedimentary basin. For each location, the central (circles) and the coincident loop (triangles) TEM data are shown together with the high-frequency MT xy (cross symbols) and yx (times symbols) data for comparison. (The data for the station located between PNB16 and PNB18 are shown in Figure 1).

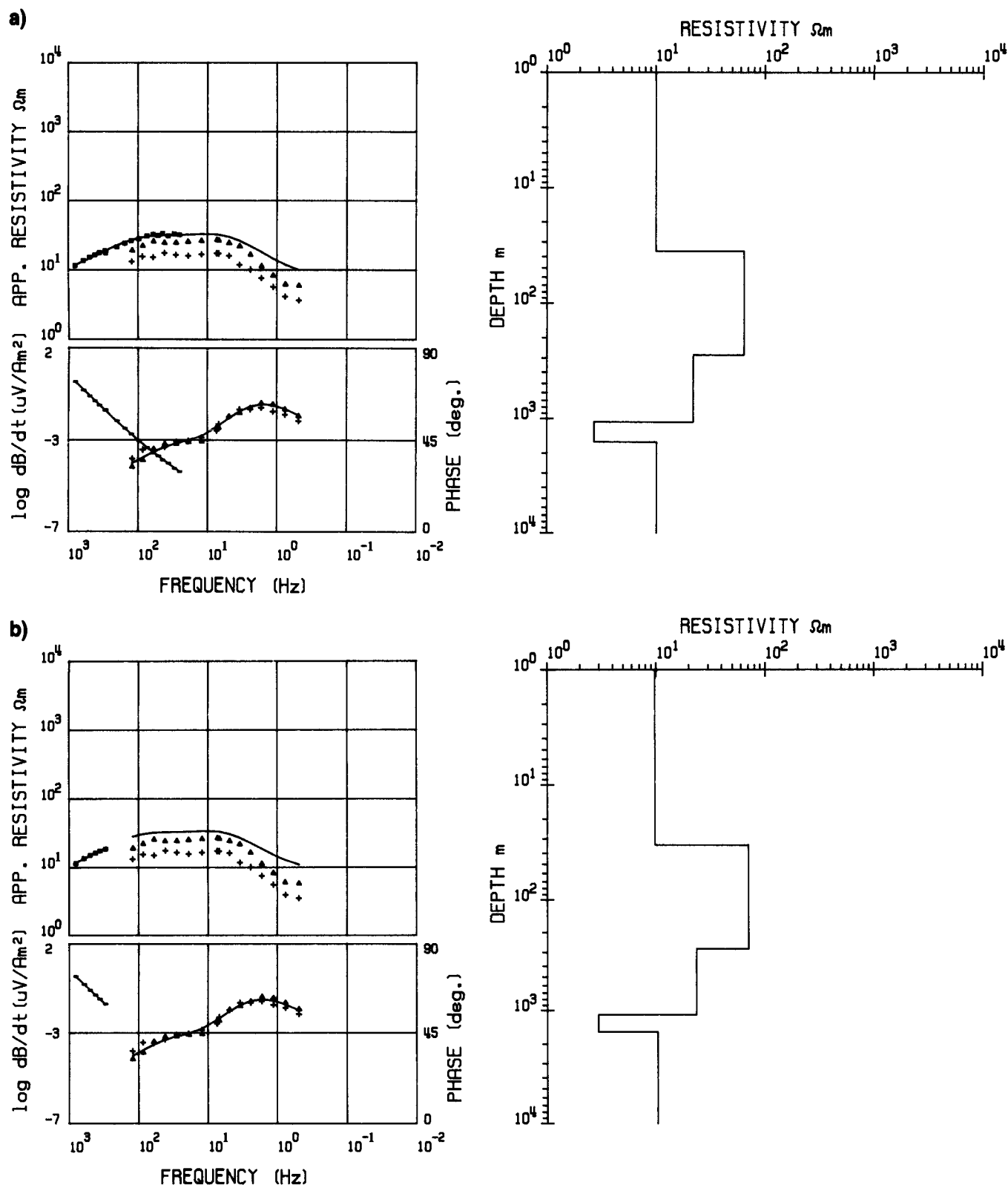


FIG. 6. Result of joint inversion of TEM apparent resistivities and MT phase data for two orthogonal directions. The MT apparent resistivity curves have been affected by different amounts of static shift. In (a), all the recorded TEM data are used in the inversion while in (b) only a nonoverlapping segment of the TEM data is considered in the inversion process. The observational errors have not been plotted in the left-hand plots for notational simplicity but were accounted for in the inversion.

static shift remediation operations. Using only the high-frequency MT data acquired over the Klirou ore body (see Figure 4), joint 1-D inversion of central loop TEM data and the rotated (major and minor) phase data were undertaken to gauge their possible usefulness in target definition. The result for the major impedance phase (whose apparent resistivity curve is similar in shape to that of TEM at the overlap region) is shown in Figure 7 and proved to be more diagnostic than the minor data (which are dissimilar to TEM data). The known position of the ore body appears to be marked by a low resistivity zone at the 80 m to 110 m depth position. The same conclusions were also reached for the comparable set of unrotated MT phase data (see Figure 4). From the foregoing naive analysis, it would appear that some useful information can be gleaned from the joint inversion of TEM data and MT phase data for matching sounding curves over a simple 3-D structure, but a proper multidimensional modeling strategy is recommended.

CONCLUSIONS

The problem of retrieving useful subsurface parameters from statically shifted MT apparent resistivity sounding curves using TEM constraints has been addressed by several workers. However, these previous studies have focused on correcting the apparent resistivity curves to some level gauged from the TEM data before carrying out an interpretation of the MT data, and such a process constitutes a two-stage interpretation scheme, *sensu stricto*. The one-

stage interpretation scheme presented here is somewhat simpler and more efficient as the practical limitations in the two-stage process (such as the requirement for an overlap region between the MT and TEM data, flat high-frequency ends of the MT curves, and the shifting of the apparent resistivity curves by an unknown factor) are reduced when dealing with data from simple geological environments. The same methods may be judiciously applied to data from complex geologies using the appropriate forward models. A naive attempt has been made to explain some consistent observations on the nature of TEM field data and their relationship with MT data. It is also stressed that TEM data acquired in the central or coincident loop geometry can be equally effective in static shift remediation operations. Moreover, since static shift only changes the level of the MT apparent resistivity curve, a few extraneous TEM data may suffice for a correction scheme in certain situations. This would help limit the amount of redundant information collected in the field and the complexity of the complimentary system deployed in the field. This point is also important in the joint inversion process as TEM forward computations are more demanding than those of MT. All the practical strategies mentioned in this paper are currently being implemented in a 2-D MT inversion scheme.

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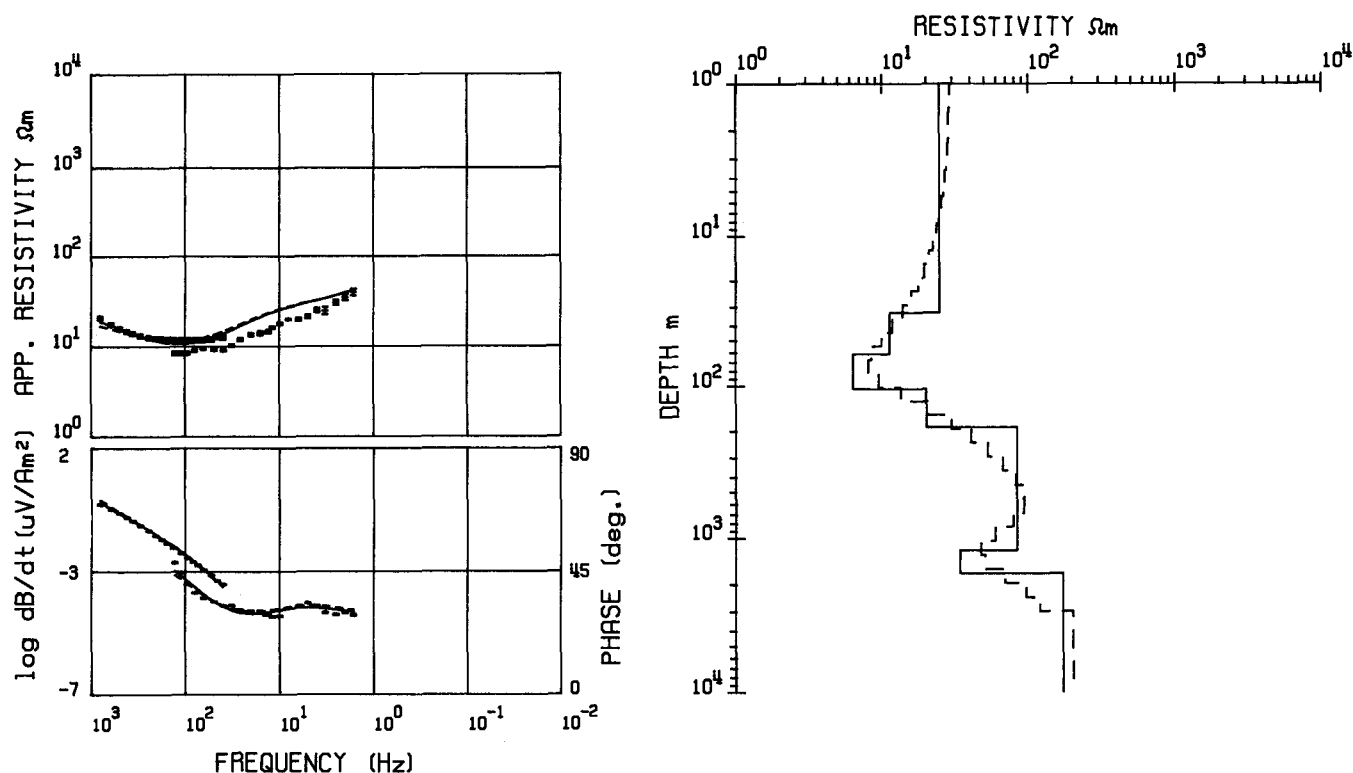


FIG. 7. Result of joint inversion of TEM and MT phase data over a known massive sulphide deposit. The smooth, densely parameterized model (shown by broken lines on the right-hand plot) was first derived from an initial half-space model of 30 Ωm resistivity. The main features of this optimal smooth model are then explored in the other sparsely parameterized optimal model (solid lines in right-hand plot). The model responses are superimposed on the field data (shown with the associated standard errors) on the left-hand plots.

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